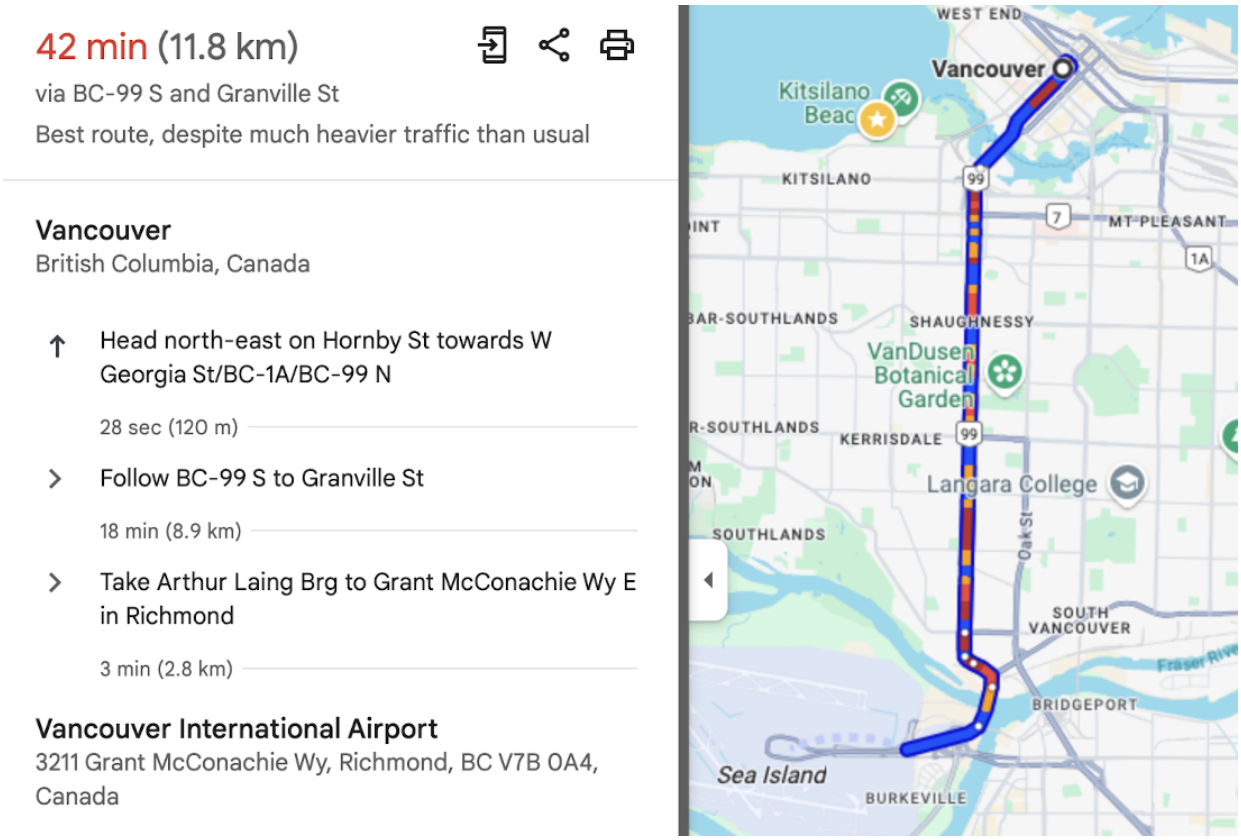


Figure 1.8 An example of a predictive digital twin provided by Google Maps showing predicted travel time. Map data © 2025 Google.

Under the hood, Google uses a Graph Neural Network (GNN) that operates on the road network, treated as a graph, as illustrated in figure 1.6. It blends live traffic data with historical trends to simulate how congestion might evolve. Google Maps combines a static representation of the physical road infrastructure with real time positional data and models trained on historical data to achieve remarkably accurate travel time predictions. The accuracy of these models became apparent during the Covid 19 pandemic when lockdowns shifted traffic patterns overnight, necessitating Google to urgently retrain them on shorter historical traffic data to retain their accuracy.